

Linear Algebra – Problem Set 1

Determinants, Cofactors & Adjugate, Cramer's Rule · Full solutions · Original problems in the spirit of MIT 18.06 (Strang)

Problem 1

A and B are 4×4 matrices with $\det A = 2$ and $\det B = -3$. Compute each of the following without finding any entries of A or B : (a) $\det(AB)$ (b) $\det(A^T)$ (c) $\det(A^{-1}B)$ (d) $\det(2A)$ (e) $\det(A^2B^{-1})$.

(a) $\det(AB) = \det A \cdot \det B = (2)(-3) = -6$

(b) $\det(A^T) = \det A = 2$

(c) $\det(A^{-1}B) = \det(A^{-1}) \det B = \frac{1}{\det A} \det B = \frac{-3}{2}$

(d) $\det(2A) = 2^4 \det A = 16 \cdot 2 = 32$ (*multiplying an $n \times n$ matrix by c multiplies each of the n rows, giving factor c^n ; here $n = 4$)*

(e) $\det(A^2B^{-1}) = (\det A)^2 \cdot \frac{1}{\det B} = 2^2 \cdot \frac{1}{-3} = -\frac{4}{3}$

Answers: (a) -6 (b) 2 (c) $-\frac{3}{2}$ (d) 32 (e) $-\frac{4}{3}$

Problem 2

A is a 3×3 matrix with $\det A = -4$. A matrix B is obtained from A by performing, in order: $R_2 \leftarrow 3R_2$, then $R_1 \leftrightarrow R_2$, then $R_3 \leftarrow R_3 - 5R_1$. Find $\det B$ and explain the effect of each operation.

Track the determinant through each operation:

- $R_2 \leftarrow 3R_2$: multiplying one row by 3 multiplies the determinant by 3. Det: $-4 \rightarrow -12$.
- $R_1 \leftrightarrow R_2$: a row swap flips the sign. Det: $-12 \rightarrow 12$.
- $R_3 \leftarrow R_3 - 5R_1$: adding a multiple of one row to *another* row is a row-replacement operation; it never changes the determinant. Det: $12 \rightarrow 12$.

$\det B = 12$

Problem 3

Starting from a 3×3 matrix A , the operations $R_2 \leftarrow 2R_2$, then $R_1 \leftrightarrow R_3$, then $R_3 \leftarrow R_3 + 4R_2$ are performed, producing

$$U = \begin{bmatrix} 3 & 1 & 2 \\ 0 & -1 & 5 \\ 0 & 0 & 4 \end{bmatrix}.$$

Find $\det A$ without reconstructing A .

U is upper triangular, so $\det U = 3 \cdot (-1) \cdot 4 = -12$.

Each operation multiplies the determinant by a known factor:

- $R_2 \leftarrow 2R_2$: factor 2

- $R_1 \leftrightarrow R_3$: factor -1
- $R_3 \leftarrow R_3 + 4R_2$: factor 1

So $\det U = (2)(-1)(1) \det A = -2 \det A$. Therefore $-12 = -2 \det A$.

$\det A = 6$

Problem 4

True or false? Justify each answer (if false, give a counterexample or correct the statement).

(a) “ $\det(A + B) = \det A + \det B$.” False. Take $A = B = I_2$: then $\det(A + B) = \det(2I_2) = 4$ but $\det A + \det B = 1 + 1 = 2$. In general the determinant is multiplicative, not additive.

(b) “If A has two equal rows, then $\det A = 0$.” True. Swapping those two rows leaves A unchanged, but a swap must flip the sign: $\det A = -\det A$, so $\det A = 0$.

(c) “If $\det(AB) = 0$, then $\det A = 0$ or $\det B = 0$.” True. By the product rule $0 = \det(AB) = \det A \cdot \det B$, and a product of real numbers is zero only when a factor is zero.

(d) “A triangular matrix whose diagonal entries are all nonzero is invertible.” True. Its determinant is the product of the diagonal entries, which is nonzero, so the matrix is invertible.

(e) “ $\det(-A) = -\det A$ for every square matrix A .” False. $\det(-A) = (-1)^n \det A$. The claim holds only for odd n ; for even n , $\det(-A) = \det A$ (e.g. $\det(-I_2) = 1 \neq -1$).

Problem 5

Let $A_t = \begin{bmatrix} t & 1 & 1 \\ 1 & t & 1 \\ 1 & 1 & t \end{bmatrix}$. Using determinant properties and row operations (not a full cofactor expansion), find $\det(A_t)$. For which values of t is A_t invertible?

All three rows sum to the same thing, so add rows 2 and 3 into row 1 (row replacement, determinant unchanged):

$$R_1 \leftarrow R_1 + R_2 + R_3 : \begin{bmatrix} t+2 & t+2 & t+2 \\ 1 & t & 1 \\ 1 & 1 & t \end{bmatrix}$$

Factor $t+2$ out of row 1:

$$\det(A_t) = (t+2) \begin{vmatrix} 1 & 1 & 1 \\ 1 & t & 1 \\ 1 & 1 & t \end{vmatrix}$$

Now clear column 1 below the pivot ($R_2 \leftarrow R_2 - R_1$, $R_3 \leftarrow R_3 - R_1$, both replacements):

$$\det(A_t) = (t+2) \begin{vmatrix} 1 & 1 & 1 \\ 0 & t-1 & 0 \\ 0 & 0 & t-1 \end{vmatrix} = (t+2)(t-1)^2.$$

$\det(A_t) = (t+2)(t-1)^2$; A_t is invertible exactly when $t \neq 1$ and $t \neq -2$.

Problem 6

Compute $\det \begin{bmatrix} 1 & 2 & 0 & 2 \\ 2 & 5 & 1 & 4 \\ 0 & 1 & 2 & 1 \\ 1 & 3 & 1 & 3 \end{bmatrix}$ by row-replacement operations only (no swaps), reducing to upper triangular form.

$$\begin{bmatrix} 1 & 2 & 0 & 2 \\ 2 & 5 & 1 & 4 \\ 0 & 1 & 2 & 1 \\ 1 & 3 & 1 & 3 \end{bmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 0 & 2 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

$$\xrightarrow[R_4 \leftarrow R_4 - R_2]{R_3 \leftarrow R_3 - R_2} \begin{bmatrix} 1 & 2 & 0 & 2 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The result is upper triangular; all operations were row replacements, so the determinant is unchanged.
 $\det = 1 \cdot 1 \cdot 1 \cdot 1 = 1$

Problem 7

Compute $\det \begin{bmatrix} 2 & 0 & 1 & 0 \\ 0 & 4 & 0 & 0 \\ 1 & 0 & 3 & 0 \\ 0 & 5 & 0 & 2 \end{bmatrix}$ by cofactor expansion, choosing the most efficient row or column and stating your choice.

Choice: expand along **column 2** — it has three zeros and a single entry 4 at position (2,2).

$$\det = 4 \cdot C_{22} = 4 \cdot (+1) \begin{vmatrix} 2 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 2 \end{vmatrix}$$

The 3×3 minor: expand along its column 3 (single entry 2 at (3,3)):

$$\begin{vmatrix} 2 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 2 \end{vmatrix} = 2 \begin{vmatrix} 2 & 1 \\ 1 & 3 \end{vmatrix} = 2(6 - 1) = 10.$$

$$\det = 4 \cdot 10 = 40$$

Problem 8

Let $A = \begin{bmatrix} 2 & 1 & 4 \\ 3 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix}$. Find the cofactors C_{21}, C_{22}, C_{23} , then compute $\det A$ by expanding along the second row. Explain the difference between a minor M_{ij} and a cofactor C_{ij} .

The cofactor is the signed minor: $C_{ij} = (-1)^{i+j} M_{ij}$.

$$C_{21} = (-1)^{2+1} \begin{vmatrix} 1 & 4 \\ 2 & 3 \end{vmatrix} = -(3-8) = 5, \quad C_{22} = (+1) \begin{vmatrix} 2 & 4 \\ 1 & 3 \end{vmatrix} = 6-4 = 2,$$

$$C_{23} = (-1)^{2+3} \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} = -(4-1) = -3.$$

Expand along row 2 (entries 3, 0, -1):

$$\det A = 3C_{21} + 0C_{22} + (-1)C_{23} = 3(5) + 0 + (-1)(-3) = 18.$$

$$C_{21} = 5, \quad C_{22} = 2, \quad C_{23} = -3; \quad \det A = 18$$

Problem 9

Let $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 2 & 0 & 1 \end{bmatrix}$. (a) Find the cofactor matrix C and $\text{adj}(A)$. (b) Verify one complete row-by-column product in $A \text{adj}(A)$. (c) Find A^{-1} .

(a) The nine cofactors:

$$C = \begin{bmatrix} (1-0) & -(0-2) & (0-2) \\ -(2-0) & (1-0) & -(0-4) \\ (2-0) & -(1-0) & (1-0) \end{bmatrix} = \begin{bmatrix} 1 & 2 & -2 \\ -2 & 1 & 4 \\ 2 & -1 & 1 \end{bmatrix}$$

Transposing gives the adjugate:

$$\text{adj}(A) = C^T = \begin{bmatrix} 1 & -2 & 2 \\ 2 & 1 & -1 \\ -2 & 4 & 1 \end{bmatrix}$$

(b) Row 1 of A times the columns of $\text{adj}(A)$:

$$(1)(1) + (2)(2) + (0)(-2) = 5, \quad (1)(-2) + (2)(1) + (0)(4) = 0, \quad (1)(2) + (2)(-1) + (0)(1) = 0.$$

Row 1 of $A \text{adj}(A)$ is $(5, 0, 0)$ — exactly $\det(A) \times (\text{row 1 of } I)$, consistent with $A \text{adj}(A) = \det(A)I$. (The other two rows check out the same way.)

(c) $\det A = 1(1-0) - 2(0-2) + 0 = 1 + 4 = 5$, so

$$A^{-1} = \frac{1}{\det A} \text{adj}(A) = \frac{1}{5} \begin{bmatrix} 1 & -2 & 2 \\ 2 & 1 & -1 \\ -2 & 4 & 1 \end{bmatrix}.$$

$$\text{adj}(A) = \frac{1}{5} \begin{bmatrix} 1 & -2 & 2 \\ 2 & 1 & -1 \\ -2 & 4 & 1 \end{bmatrix}, \quad \det A = 5, \quad A^{-1} = \frac{1}{5} \text{adj}(A)$$

Problem 10

Use Cramer's Rule to solve $\begin{cases} 3x - 2y = 4 \\ 2x + 5y = 9 \end{cases}$. Identify A , A_1 , A_2 and compute the necessary determinants.

$$A = \begin{bmatrix} 3 & -2 \\ 2 & 5 \end{bmatrix}, \quad \det A = 15 + 4 = 19 \neq 0 \checkmark \text{ (Cramer applies)}$$

$$A_1 = \begin{bmatrix} 4 & -2 \\ 9 & 5 \end{bmatrix}, \quad \det A_1 = 20 + 18 = 38; \quad A_2 = \begin{bmatrix} 3 & 4 \\ 2 & 9 \end{bmatrix}, \quad \det A_2 = 27 - 8 = 19.$$

$$x = \frac{\det A_1}{\det A} = \frac{38}{19} = 2, \quad y = \frac{\det A_2}{\det A} = \frac{19}{19} = 1.$$

$x = 2, y = 1$ (**check:** $3(2) - 2(1) = 4$, $2(2) + 5(1) = 9$, **both hold**)

Problem 11

Consider the system

$$\begin{cases} x + 2y + z = 5 \\ 2x - y + 3z = 4 \\ x + y + 2z = 7 \end{cases}$$

Use Cramer's Rule to find *only* y . Do not solve for x or z .

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 3 \\ 1 & 1 & 2 \end{bmatrix}, \quad \det A = 1(-2 - 3) - 2(4 - 3) + 1(2 + 1) = -5 - 2 + 3 = -4.$$

Replace column 2 by $b = (5, 4, 7)$:

$$A_2 = \begin{bmatrix} 1 & 5 & 1 \\ 2 & 4 & 3 \\ 1 & 7 & 2 \end{bmatrix}, \quad \det A_2 = 1(8 - 21) - 5(4 - 3) + 1(14 - 4) = -13 - 5 + 10 = -8.$$

$$y = \frac{\det A_2}{\det A} = \frac{-8}{-4} = 2.$$

$y = 2$ (**indeed the full solution is** $x = -3, y = 2, z = 4$, **but Cramer let us get y without touching x, z**)

Problem 12

- (a) Suppose every row of a square matrix A sums to zero. Prove that $\det A = 0$ without expanding the determinant. (b) True or false: “ $\text{adj}(A)$ exists only when A is invertible.” Justify your answer.

(a) Proof. Let $\mathbf{1} = (1, 1, \dots, 1)^T$. Since every row of A sums to zero, each entry of $A\mathbf{1}$ is a row-sum, so

$$A\mathbf{1} = \mathbf{0}.$$

Thus $A\mathbf{x} = \mathbf{0}$ has a nonzero solution ($\mathbf{1} \neq \mathbf{0}$), so A is not invertible, hence $\det A = 0$. ■

Alternative proof: adding columns $2, \dots, n$ into column 1 (row-replacement in disguise, determinant unchanged) produces a zero column, and a matrix with a zero column has determinant 0.

(b) False. The adjugate is built from cofactors, which exist for *every* square matrix, invertible or not. What is always true is the identity $A \operatorname{adj}(A) = \det(A) I$ — when $\det A = 0$ the right side is just the zero matrix. Example:

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}, \quad \det A = 0, \quad \operatorname{adj}(A) = \begin{bmatrix} 4 & -2 \\ -2 & 1 \end{bmatrix} \neq 0,$$

and indeed $A \operatorname{adj}(A) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0 \cdot I$. So a singular matrix can have a nonzero adjugate.

(a) $A\mathbf{1} = \mathbf{0}$ forces $\det A = 0$. **(b) False** — $\operatorname{adj}(A)$ always exists; singularity only kills $\det A$, not the cofactors.